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Extracted Text

LY R [4329] **L Y R**

Lyre, which he ufd to frequent, at the foot of a mountain in Arcadill. In all as he was going in, he found a tor- tulofe finding at the entrance of his case: he killed him poor creature, and perhaps, vat the flesh of it. As he was diverting himself with the flesh, he was migliedly pleafed with the molle of great from his concave figure. He had puffily been cunning enough to find one, that a thong pulled fruit and affected is each end, when freds by things you take aird mudiculous. How ever that was, he went immediately to work, and eut feveral things out of the links he had lately flolen, and afected them as right as we could on the field of the tortofie and, in playing with them, made a new kind of mute with them to divert himself in his retreat. This, considered only as an account of the first inven- tion of the lyre, is not altogether for unsurals?

The most ancient representations of this instrument ages very well with the account of its uncertain the lyre, in particular on the old echelith globes, was re- prefered as made of one entire shell of a tortorise; and that, Amphibus in the celebrated group of the Diver, or Toro, in the Fanerle place at Rome, which is of Greek feulpture, and very high authority, is figured in the fine manner. See Plate EXXL fig. 7.

There have, however, been many other claimants to the fewven-firinged lyre. For though Mercury invented the thin instruments in the manner already related, it's fail for afterwards gave it to a position, who are the first that played upon it with method, and made it the con- fiant companion of pretty. According to Moment ac scount of this transfections, in his hymn to Aftervery, it was given by that god to Apollo, as a peace offering and indemnification for the oxen which he had flolen from him:

Tv Phachos Maiss fun preints the lyre, A A just intended to aproils like it.

A post contains it aproils, and clays. The need information a threshold was). With exact least full the platform readily, and fines. Most seen according to the resulting fringes. Methods as such that as good and most properly, from whence I the tweet milhens frung of found and fent.

Doulton informs us, that Apollo four reporting of the creety with which he had treated Marxas in confidence of their unified control, broke the fringe of the lyre, and by that means put a flop for a time to any further progress in the practice of that new in frumers. "The music (aida by afterwards added to this infrument the firing called my/y Linas, that of _Ikshary_) and Orpheus Thanayas, hide finds which are caused _Update and participate_ (i).

Again, many ancient and repeticable authors tell

us, that, before the time of Terpandan, the Grecian lyre had only four fingers and; if so may believe Biduan, is remained in alien face 850 years, from the time of Amphio, till Terpander added to it three new fringes, which extended the mulleical feel to a hope, chord, or events, and luppiled the player with two conjoint tetracothis. It was about 150 years after this period, that Pythogorsa is fast to have added an eighth fringe to the lyre, in order to complete the oc- rave, which conflicted of two disjunctiont herchords.

Bochthia gives a different history of the feels, and tells me, that the flyton did no long remains in fact narrow limits as a tetrachold. Chorobus, the fan of Athis, or Arys, king of Lydia, added a fifth firing; Hyguns,

a firstly, Terpandera, a favorite and, at large), Erythema of Samoa, an eighth. But all their accounts are irreconcileable with Momars, hymn to Mercury, where the chelys, or reduce, the invention of which he earlier or than god, is fast to have had been fittings. There are many clamans among the musicians of ancient Greece, to the things that were afterwards added to thefly, by which the date, in the time of Alfoxcoras, was extended to two offices. Athamus, more than once, peaks of the nine fingerling inflamments, and Lon OlChoja rtagies and lyrics pace and philosopher, who fitf recited his pieces in the 8zd olympiad, 452 B. C. mentions, in some refer quoted in Dr. David, the ten- firginged lyrs, a prodt that, the third conjoint tetrchord was added to the scale in his time, which was about 50 years after Pythagoras is fuppelled to have confinational the death.

The different claimants among the Grecks to the famedual differences, only proxy, that made was cultreated in different countries, and thus the inhabitants of each country invented and improved their own in-fruments, none of which happening to celebrate those of other parts of Greece, resident it difficult for his- florions to avoid attributing the fame invention to dif- ferent periods. Thus the Eagle Rule we give to Mi- nerva, and to Marfayas theynas, or filialty, to Pan, and to Cybele, and the lyre, or citura, to Mercury, Apollo, Amphioa, Lima and Opheus. Indeed, the mere addition of a living or now to an infiltration with on a neck, was fo obvious and early, that it is scarce podible not to conceive many people to have done it at the fame time.

With respect to the form of the ancient lyre; as little agreement is to be found among authors as about the number of things. The belt endurance concerning it are the representations of this inflamment in the hands of ancient flatues, bas-reliefs, etc. See Plate CLXI.

Lightning, we have few always; it is _mattematically couldnt_, that, when a _returning firele_ comes to be produced by the _fuddro_ removal of the very frong elafic electrical prefi-fore of the electrical atmosphere of a thunder-cloud powerfully charged; the frength of such a retraining _fireks_ mult be enormous.

" If indeed the quantity of electric field naturally contained in the body of a man, forrindance, were often mefe, or induding, the author's ultimate between the effects producible by a cloud, and those caned by a prime conductor might be admitted. But fairly electronic cloud,how great forces may be its extent, and the height of its charge, when compared with the scenes and change of a prime conductor– cannot cepel from a mared (way (only other key)) more than the natural quantity of electricity which it contains. On the fodder removal, therefore of the preffure by which this natural quantity had been expelled, in consequence of the equation of the cloud into the earth) so more (at the utmost) than his _robole natural lack_ of electricity can re-enter the body (c). But we have no reason to fopate that this quantity is fo great, as that its fudden reentrance into his body found delivery on even injury third " At the experiment above deferbed, the infulated perfor receives into his body, at the infant of the _returning firele_ has only all that portion of his own natural electric-free which had been expelled from it; but likewise transmits through it, at the fame infant, _its consequences of the peculiar fluctuation_, all the electric fire of which the large cloud could had been robbeds and which mult needeffally re-pafa through his body, to arrive at that conductor. To render the cafe foreshable instability in natural electricity, the man's body mult be fo peculiarly circumdanced, impunding him to be in a hoofu, that the electric matter which has been expelled from the head into the earth, by the preffure of an extensive thundercloud, could not return back into the building, on the explorations of the clouds as different, without paling through his body: a safe not likely to happen, unfuls the boots were infolated (like the second conductor in the preceding experiments), and his body became the claimed through which where the hole could have the electric matter reflored to it : it appears much more probable that the electric matter returns to the house through the _fame elements_ by whole it before recently pafled out, and with equal silence, too more fuddently.

" If the cafe of a man who is abroad, and in an open field, during the time of an explosion a he is unconnected with other matter of matter above him, no more than the preflic quantity of electric fire, which had been before expelled from his body, will definitely return into it at the infant of a dither explosion; and that this quantity is not uufaly very large, may be inferred from many considerations.

" When a perfon standing on the ground holds a Lightning pair of Mr Canton's balls in his hand, while a highly charged thunder-cloud is fufpended over his head; the angle made by the balla indicates the _electrical fate_ of this portion or the quantity of natural electricity of which his body as at the time deprived, by the action of the (_politively_) charged cloud hanging over him. But we have never feen the repollion of the balls to fulfillment, as to contain any the apprehensions that the return of his natural electric matter, however fudden, could be attached with his injury to him a nor would be be flexible of any common use

the bulk fuddily coming together; the 'a fpark might un-doubtedly be perceived, at that infants, were he infuried, and placed in the four manner with the author when he tried the above-related experiment.

" The author nevertheless observe, that , there have been influences of protons who have been killed by natural electricity, having been found with their _floos torna_, and with their feet damaged by the electrical line, but who have not bad, over their whole boo, any other apparent marks of having been ftuck with lightning." He adds, " if a man walking out of doors were to be killed by a returning firele, the electrical line would rub into that each body into his _feets_, and his feet only ; which would not be the cafe, here he to be killed by any _main fireke_ of exploding, either positive negative.

" It would be no difficult talk, we think, to account for these appearances in different manner, were all the circumstances attending the cage minutely after-tained, but without interrogating the dead on this futjels, we may more falsifiedly appeal to the ex-perience of the living (a), to flew, that though the _returning fireks_ mult take place, in all thunder-forms, _in June degree or other_; yet it is now of that alarming magnitude within the author afterhes to it. If, a non particular thunder-form, a man in the open fields could be killed, at the infant of a dillank explotion, merely by the returners of his new electric fire, which had before been driven out of his body; lureely numerous obferations of persons who had experienced the _returning firele_, in fighting degrees, would be families, and favors a great teacher-theme must have occurred, in which one perform or another mult not, at theinflated an explosion, have felt the effects of the _returning firele_, in _foon degree or other_ from that of a violent sanctions, to that of a flight and almost impregnible pollution. But no observances we this kind are known to us say how we ever heard of any person's experiencing any kind of electrical com- motion in a thunder-loria, except fish as have either been directly frocks, or have happened to be in the _royar near neighborhood_ of the foot where the explodes took place.

" The author has been aware of this objections, which (c) " We fuppie the perfor not to be fo fluuted, that the returning fire of other bodies mult necefarily pafa through his body.

(b) " The author does indeed produce a living evidence, in the cafe of a person at Vienna, who, he has been credibly informed, received an electrical shock, by having held one hand accidentally in contact with an intercepted metallic coaching mind, at the infant that a thunder-cloud exploited at the influence, as was considered, or above half an English milk. He likewise observes, that a very strong, bright, and fulden firek (ce fpars) of electric fire has been been, by levels electrificants, to pais in the internal, or interruption, purposefully left in the concluding rod of a book at the infant do a distant explosions, and when it was fully proved, by the sharp point of the conductor not being melted, or even tuged,' that the conductor itself had not been effect "Other observations, however, do not by any means prove the magnitude or danger, of the returning broke, but merely its _esiflessen_ which we do not contiti."

Source record

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Lyre. which he uſed to frequent, at the foot of a mountain in Arcadia. Juſt as he was going in, he found a tortoiſe feeding at the entrance of his cave: he killed the poor creature, and, perhaps, eat the fleſh of it. As he was diverting himſelf with the ſhell, he was mightily pleaſed with the noiſe it gave from its concave figure. He had poſſibly been cunning enough to find out, that a thong pulled ſtrait and faſtened at each end, when ſtruck by the finger, made a ſort of muſical ſound. However that was, he went immediately to work, and cut ſeveral thongs out of the hides he had lately ſtolen, and faſtened them as tight as he could to the ſhell of this tortoiſe; and, in playing with them, made a new kind of muſic with them to divert himſelf in his retreat. This, conſidered only as an account of the firſt invention of the lyre, is not altogether ſo unnatural."

The moſt ancient repreſentations of this inſtrument agree very well with the account of its invention: the lyre, in particular on the old celeſtial globes, was repreſented as made of one entire ſhell of a tortoiſe; and that Amphion in the celebrated group of the Dirce, or Toro, in the Farneſe palace at Rome, which is of Greek ſculpture, and very high antiquity, is figured in the ſame manner. See Plate CLXI. fig. 9.

There have, however, been many other claimants to the ſeven-ſtringed lyre. For though Mercury invented this inſtrument in the manner already related, it is ſaid he afterwards gave it to Apollo, who was the firſt that played upon it with method, and made it the conſtant companion of poetry. According to Homer's account of this tranſaction, in his hymn to Mercury, it was given by that god to Apollo, as a peace-offering and indemnification for the oxen which he had ſtolen from him:

> To Phœbus Maia's ſon preſents the lyre,
> A gift intended to appeaſe his ire.
> The god receives it gladly, and eſſays
> The novel inſtrument a thouſand ways:
> With dext'rous ſkill the plectrum wields; and ſings,
> With voice accordant to the trembling ſtrings,
> Such ſtrains as gods and men approv'd, from whence
> The ſweet alliance ſprung of ſound and ſenſe.

Diodorus informs us, that Apollo ſoon repenting of the cruelty with which he had treated Marsyas in conſequence of their muſical conteſt, broke the ſtrings of the lyre, and by that means put a ſtop for a time to any further progreſs in the practice of that new inſtrument. "The muſes (adds he) afterwards added to this inſtrument the ſtring called *meſe;* Linus, that of *lichanos;* and Orpheus and Thamyras, thoſe ſtrings which are named *hypate* and *parhypate* (A).

Again, many ancient and reſpectable authors tell us, that, before the time of Terpander, the Grecian lyre had only four ſtrings; and, if we may believe Suidas, it remained in this ſtate 856 years, from the time of Amphion, till Terpander added to it three new ſtrings, which extended the muſical ſcale to a heptachord, or ſeventh, and ſupplied the player with two conjoint tetrachords. It was about 150 years after this period, that Pythagoras is ſaid to have added an eighth ſtring to the lyre, in order to complete the octave, which conſiſted of two disjunct tetrachords. Lyre.

Boethius gives a different hiſtory of the ſcale, and tells us, that the ſyſtem did not long remain in ſuch narrow limits as a tetrachord. Chorœbus, the ſon of Athis, or Atys, king of Lydia, added a fifth ſtring; Hyagnis, a ſixth; Terpander, a ſeventh; and, at length, Lychaon of Samos, an eighth. But all theſe accounts are irreconcileable with Homer's hymn to Mercury, where the chelys, or teſtudo, the invention of which he aſcribes to that god, is ſaid to have had ſeven ſtrings. There are many claimants among the muſicians of ancient Greece, to the ſtrings that were afterwards added to theſe, by which the ſcale, in the time of Ariſtoxenus, was extended to two octaves. Athenæus, more than once, ſpeaks of the nine-ſtringed inſtrument; and Ion of Chios, a tragic and lyric poet and philoſopher, who firſt recited his pieces in the 82d olympiad, 452 B. C. mentions, in ſome verſes quoted by Euclid, the ten-ſtringed lyre; a proof that the third conjoint tetrachord was added to the ſcale in his time, which was about 50 years after Pythagoras is ſuppoſed to have conſtructed the octachord.

The different claimants among the Greeks to the ſame muſical diſcoveries, only prove, that muſic was cultivated in different countries; and that the inhabitants of each country invented and improved their own inſtruments, ſome of which happening to reſemble thoſe of other parts of Greece, rendered it difficult for hiſtorians to avoid attributing the ſame invention to different perſons. Thus the ſingle flute was given to Minerva, and to Marſyas; the ſyrinx, or fiſtula, to Pan, and to Cybele; and the lyre, or cithara, to Mercury, Apollo, Amphion, Linus, and Orpheus. Indeed, the mere addition of a ſtring or two to an inſtrument without a neck, was ſo obvious and eaſy, that it is ſcarce poſſible not to conceive many people to have done it at the ſame time.

With reſpect to the form of the ancient lyre; as little agreement is to be found among authors as about the number of ſtrings. The beſt evidences concerning it are the repreſentations of that inſtrument in the hands of ancient ſtatues, bas-reliefs, &c. See Plate CLXI.

(A) It has been already related, that the lyre invented by the Egyptian Mercury had but three ſtrings; and by putting theſe two circumſtances together, Dr Burney obſerves, we may perhaps acquire ſome knowledge of the progreſs of muſic, or, at leaſt, of the extenſion of its ſcale, in the higheſt antiquity.

Meſe, in the Greek muſic, is the fourth ſound of the ſecond tetrachord of the great ſyſtem, and firſt tetrachord invented by the ancients, anſwering to our A, on the fifth line in the baſe. If this ſound then was added to the former three, it proves two important points: firſt, that the moſt ancient tetrachord was that from E in the baſe to A; and that the three original ſtrings in the Mercurian and Apollonian lyre were tuned E, F, G, which the Greeks called *Hypate Meſon, Parhypate Meſon, Meſon Diatonos*. The addition therefore of *Meſe* to theſe, completed the firſt and moſt ancient tetrachord, E, F, G, A.

The ſtring *lichanos*, then, being added to theſe, and anſwering to our D on the third line in the baſe, extended the compaſs downwards, and gave the ancient lyre a regular ſeries of five ſounds in the Dorian mode, the moſt ancient of all the Greek modes; and the two ſtrings called *Hypate* and *Parhypate*, correſponding with our B and C in the baſe, completed the heptachord, or ſeven ſounds, B, C, D, E, F, G, A, a compaſs that received no addition till after the time of Pindar, who calls the inſtrument then in uſe the *ſeven-tongued lyre*.

Figure 1: *
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Lightning. we have seen above; it is *mathematically evident*, that, when a *returning ſtroke* comes to be produced by the *ſudden* removal of the very ſtrong elaſtic electrical preſſure of the electrical atmoſphere of a thunder-cloud powerfully charged; the ſtrength of ſuch a *returning ſtroke* muſt be *enormous*.'

" If indeed the quantity of electric fluid *naturally* contained in the body of a man, for inſtance, were immenſe, or indefinite, the author's eſtimate between the effects producible by a cloud, and thoſe cauſed by a prime conductor, might be admitted. But ſurely an electrified cloud,—how great ſoever may be its extent, and the height of its charge, when compared with the extent and charge of a prime conductor—cannot expel from a man's body (or any other body) *more* than the *natural* quantity of electricity which it contains. On the ſudden removal, therefore, of the preſſure by which this natural quantity had been expelled, in conſequence of the exploſion of the cloud into the earth; *no more* (at the utmoſt) than his *whole natural ſtock* of electricity can re-enter his body (c). But we have no reaſon to ſuppoſe that this quantity is ſo great, as that its ſudden re-entrance into his body ſhould deſtroy or even injure him.

" In the experiment above deſcribed, the inſulated perſon receives into his body, at the inſtant of the *returning ſtroke*, not only all that portion of his own natural electric-fire which had been expelled from it; but likewiſe tranſmits through it, at the ſame inſtant, *in conſequence of his peculiar ſituation*, all the electric fire of which the large ſecond conductor had been robbed; and which muſt neceſſarily *re*-paſs through his body, to arrive at that conductor. To render the caſe ſomewhat parallel, in *natural electricity*, the man's body muſt be ſo peculiarly circumſtanced, ſuppoſing him to be in a houſe, that the electric matter which has been expelled from the houſe into the earth, by the preſſure of an extenſive thunder-cloud, could not return back into the building, on the exploſion of the cloud at a diſtance, without paſſing through his body: a caſe not likely to happen, unleſs the houſe were inſulated (like the ſecond conductor in the preceding experiment), and his body became the channel through which alone the houſe could have its electric matter reſtored to it: it appears much more probable that the electric matter returns to the houſe through the *ſame channels* by which it before inſenſibly paſſed out, and with equal ſilence, tho' more ſuddenly.

" In the caſe of a man who is abroad, and in an open field, during the time of an exploſion;—as he is unconnected with other maſſes of matter *above him*, no more than the preciſe quantity of electric fire, which had been before expelled from his body, will *ſuddenly* return into it at the inſtant of a diſtant exploſion: and that this quantity is not uſually very large, may be inferred from many conſiderations.

" When a perſon ſtanding on the ground holds a Lightning. pair of Mr Canton's balls in his hand, while a highly charged thunder-cloud is ſuſpended over his head; the angle made by the balls indicates the *electrical ſtate* of that perſon, or the quantity of *natural* electricity of which his body is at that time deprived, by the action of the *(poſitively)* charged cloud hanging over him. But we have never ſeen the repulſion of the balls ſo conſiderable, as to furniſh any juſt apprehenſions that the return of his natural electric matter, however ſudden, could be attended with injury to him: nor would he be ſenſible of any commotion on the balls ſuddenly coming together; tho' a ſpark might undoubtedly be perceived, at that inſtant, were he inſulated, and placed in the ſame manner with the author when he tried the above-related experiment.

" The author nevertheleſs obſerves, that ' there have been inſtances of perſons who have been killed by *natural* electricity, having been found with their *ſhoes torn*, and with their *feet* damaged by the electrical fire; but who have not had, over their whole body, any other apparent marks of having been ſtruck with lightning.' He adds, ' if a man walking out of doors were to be killed by a *returning ſtroke*, the electrical fire would ruſh into that man's body thro' his *feet*, and his feet only; which would *not* be the caſe, were he to be killed by any *main ſtroke* of exploſion, either poſitive or negative.'

" It would be no difficult taſk, we think, to account for theſe appearances in a different manner; were all the circumſtances attending the caſe minutely aſcertained: but without interrogating the *dead* on this ſubject, we may more ſatisfactorily appeal to the experience of the *living* (d), to ſhew, that though the *returning ſtroke* muſt take place, in all thunder-ſtorms, *in ſome degree or other*; yet it is not of that alarming magnitude which the author aſcribes to it. If, in any particular thunder-ſtorm, a man in the open fields could be killed, at the inſtant of a diſtant exploſion, merely by the *return* of his *own* electric fire, which had before been driven out of his body; ſurely numerous obſervations of perſons who had experienced the *returning ſtroke*, in *ſlighter* degrees, would be familiar; and ſcarce a great thunder-ſtorm muſt have occurred, in which one perſon or another muſt not, at the inſtant of an exploſion, have felt the effects of the *returning ſtroke*, in *ſome degree or other*— from that of a violent *concuſſion*, to that of a ſlight and almoſt imperceptible *pulſation*. But no obſervations of this kind are known to us; nor have we ever heard of any perſon's experiencing any kind of electrical commotion in a thunder-ſtorm, except ſuch as have either been directly ſtruck, or have happened to be in the *very near neighbourhood* of the ſpot where the exploſion took place.

" The author has been aware of this objection, which

(c) " We ſuppoſe the perſon not to be ſo ſituated, that the *returning fire* of other bodies muſt neceſſarily paſs through his body."

(d) " The author does indeed produce a *living* evidence, in the caſe of a perſon at Vienna, who, he has been credibly informed, received an electrical ſhock, by having held one hand accidentally in contact with an *interrupted* metallic conducting rod, at the inſtant that a thunder-cloud exploded at the diſtance, as was conjectured, of above half an Engliſh mile. He likewiſe obſerves, that a ' very ſtrong, bright, and ſudden *ſtroke* ' (or ſpark) of electrical fire has been ſeen, by ſeveral electricians, to paſs in the interval, or interruption, purpoſely left in the conducting rod of a houſe at the inſtant of a diſtant exploſion; and ' when it was fully proved, by the ſharp point of the conductor not being melted, or even tinged,' that the conductor itſelf had not been ſtruck.—Theſe obſervations, however, do not by any means prove the *magnitude*, or *danger*, of the returning ſtroke, but merely its *exiſtence*: which we do not conteſt."

Figure 2: *
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